QUATERLY EXAMINATION - 2026

STD : 12

சேலம்

MATHEMATICS

MARK : 90

TIME : 3.00 HRS

PART - A

CHOOSE THE CORRET ANSWER :- 20 X 1 = 20

1) If $A\begin{bmatrix}1 & -2\\1 & 4\end{bmatrix}=\begin{bmatrix}6 & 0\\0 & 6\end{bmatrix}$, then $A =$

(a) $\begin{bmatrix}1 & -2\\1 & 4\end{bmatrix}$ (b) $\begin{bmatrix}1 & 2\\-1 & 4\end{bmatrix}$ (c) $\begin{bmatrix}4 & 2\\-1 & 1\end{bmatrix}$ (d) $\begin{bmatrix}4 & -1\\2 & 1\end{bmatrix}$

2) If $x^ay^b=e^m, x^cy^d=e^n, \Delta_1=\begin{vmatrix}m & b\\n & d\end{vmatrix}, \Delta_2=\begin{vmatrix}a & m\\c & n\end{vmatrix}, \Delta_3=\begin{vmatrix}a & b\\c & d\end{vmatrix}$, then the values of x and y are respectively,

(a) $e^{(\Delta_2/\Delta_1)}, e^{(\Delta_3/\Delta_1)}$ (b) $\log(\Delta_1/\Delta_3), \log(\Delta_2/\Delta_3)$

(c) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ (d) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$

3) The solution of the equation $|z|-z=1+2i$ is

(a) $\frac{3}{2}-2i$ (b) $-\frac{3}{2}+2i$ (c) $2-\frac{3}{2}i$ (d) $2+\frac{3}{2}i$

4) The principal argument of the complex number $\frac{(1+i\sqrt{3})^2}{4i(1-i\sqrt{3})}$ is

(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{5\pi}{6}$ (d) $\frac{\pi}{2}$

5) If α, β and γ are the roots of x^3+px^2+qx+r, then $\sum\frac{1}{\alpha}$ is

(a) $-\frac{q}{r}$ (b) $-\frac{p}{r}$ (c) $\frac{q}{r}$ (d) $-\frac{q}{p}$

6) The number of real numbers in $[0,2\pi]$ satisfying $\sin^4 x-2\sin^2 x+1$ is

(a) 2 (b) 4 (c) 1 (d) ∞

7) $\sin^{-1}\frac{3}{5}-\cos^{-1}\frac{12}{13}+\sec^{-1}\frac{5}{3}-\operatorname{cosec}^{-1}\frac{13}{12}$ is equal to

(a) 2π (b) π (c) 0 (d) $\tan^{-1}\frac{12}{65}$

8) The domain of the function defined by $f(x)=\sin^{-1}\sqrt{x-1}$ is

(a) $[1,2]$ (b) $[-1,1]$ (c) $[0,1]$ (d) $[-1,0]$

9) If $|x|\le 1$, then $2\tan^{-1}x-\sin^{-1}\frac{2x}{1+x^2}$ is equal to

(a) $\tan^{-1}x$ (b) $\sin^{-1}x$ (c) 0 (d) π

10) The centre of the circle inscribed in a square formed by the lines $x^2-8x-12=0$ and $y^2-14y+45=0$ i

(a) (4,7) (b) (7,4) (c) (9,4) (d) (4,9)

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11) Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ parallel to the straight line $2x - y = 1$. One of the points of contact of tangents on the hyperbola is

(a) $\left(\frac{9}{2\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ (b) $\left(\frac{-9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (c) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (d) $(3\sqrt{3}, -2\sqrt{2})$

12) The values of m for which the line $y = mx + 2\sqrt{5}$ touches the hyperbola $16x^2 - 9y^2 = 144$ are the roots of $x^2 - (a + b)x - 4 = 0$, then the value of $(a + b)$ is

(a) 2 (b) 4 (c) 0 (d) -2

13) If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

(a) 2 (b) -1 (c) 1 (d) 0

14) If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \frac{\pi}{4}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

15) If the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$, then (α, β) is

(a) $(-5,5)$ (b) $(-6,7)$ (c) $(5,-5)$ (d) $(6,-7)$

16) The value of is $tan\left(\frac{1}{2}cos^{-1}\left(\frac{\sqrt{5}}{3}\right)\right)$ is

(a) $\frac{5-\sqrt{3}}{2}$ (b) $\frac{5+\sqrt{3}}{2}$ (c) $\frac{3+\sqrt{5}}{2}$ (d) $\frac{3-\sqrt{5}}{2}$

17) If value of $(1+i)^4\,(1+\frac{1}{i})^4$ is

(a) 12 (b) 2 (c)8 (d) 16

18) If A is a non –zero column matrix of order m x 1 and B is a non – zero row matrix of order 1 x n then rank of AB is

(a) m (b) n (c) mn (d) 1

19) The locus of the point of intersection of perpendicular tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is

(a) $x^2 + y^2 = 25$ (b) $x^2 + y^2 = 7$ (c) $x^2 + y^2 = 3$ (d) $x^2 + y^2 = 4$

20) If $[\ \vec{a} \times \vec{b}\ ,\ \vec{b} \times \vec{c}\ ,\ \vec{c} \times \vec{a}\ ] = \lambda\left[\vec{a}, \vec{b}, \vec{c}\right]^2$ then the value of λ is

(a) 1 (b) 2 (c) 4 (d) -2

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PART – B

ANSWER ANY SEVEN QUESTIONS (Q.NO : 30 IS COMPULSORY) :- 7 X 2 = 14

21. If A is a nonsingular matrix of odd order, prove that $|\text{adj } A|$ is positive.

22. Find z^{-1}, if $z = (2 + 3i)(1 - i)$.

23. If $\omega \neq 1$ is a cube root of unity, show that $\frac{a+b\omega+c\quad 2}{b+c\omega+a\quad 2} + \frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} = -1$.

24. A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was left standing.

25. For what value of x does $\sin x = \sin^{-1} x$?

26. Find the value of $\tan(\tan^{-1}(-0.2021))$.

$c^2 = a^2(1+m^2)$

$y = mx + c$ $\quad x^2+y^2=a^2$

27. If $y = 2\sqrt{2}x + c$ is a tangent to the circle $x^2 + y^2 = 16$, find the value of c.

28. Show that the absolute value of difference of the focal distances of any point P on the hyperbola is the length of its transverse axis.

29. Find the angle between the following lines $\frac{x+4}{3} = \frac{y-7}{4} = \frac{z+5}{5}$ and $\vec{r} = 4\hat{k} + t(2\hat{i} + \hat{j} + \hat{k})$.

30. If the line $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+\lambda}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ are intersecting then find the value of λ.

PART – C

ANSWER ANY SEVEN QUESTIONS (Q.NO : 40 IS COMPULSORY) :- **7 X 3 = 21**

31. Find the rank of the matrix $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$ b\y reducing it to an echelon form.

32. Show that the equation $z^3 + 2\bar{z} = 0$ has five solutions.

33. Solve the cubic equation: $2x^3 - x^2 - 18x + 9 = 0$ if sum of two of its roots vanishes.

$4\overline{)180}$ $\quad 4$ $\quad 16$ $\quad 20$

34. Find the domain of $\cos^{-1}\left(\frac{2+\sin}{3}\right)$.

35. Find the value of $\tan^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

36. Prove that the point of intersection of the tangents at $'t_1'$ and $'t_2'$ on the parabola $y^2 = 4ax$ is $[at_1t_2, a(t_1 + t_2)]$.

37. The parabolic communications antenna has a focus at $2m$ distance from the vertex of the antenna. Find the width of the antenna $3m$ from the vertex

38. Find the torque (moment) of the resultant of the three forces represented by $\vec{\tau} = \vec{r} \times \vec{F}$
$-3\hat{i} + 6\hat{j} - 3\hat{k}, 4\hat{i} - 10\hat{j} + 12\hat{k}$ and $4\hat{i} + 7\hat{j}$ acting at the point with position vector
$8\hat{i} - 6\hat{j} - 4\hat{k}$, about the point with position vector $18\hat{i} + 3\hat{j} - 9\hat{k}$.

$\frac{\pi}{3}$ $\quad 6\overline{)180}$

39. If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a} \times (\hat{b} \times \hat{c}) = \frac{1}{2}\hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$.

$\frac{z+\lambda}{7} = \frac{z-}{5}$

$(z+\lambda)5 = 6-$

40. Show that the system of linear equation $3x - 4y = 2$, $5x + 2y = 12$, $-x + 3y = 1$ are consistent.

$5z+5\lambda = 7z-$

$-2z+5\lambda = -3$

$\frac{\pi}{6}$ $\quad 30$

PART – D

ANSWER ALL THE QUESTIONS :- **7 X 5 = 35**

41 a. A family of 3 people went out for dinner in a restaurant. The cost of two dosai, three idlies and two vada is is ₹150. The cost of the two dosai, two idlies and four vadais is ₹200. The cost of five dosai,

$\frac{2\pi}{3}$

$3\overline{)180}$

four idlies and two vadais is ₹250. The family has ₹350 in hand and they ate 3 dosai and six idlies and six vadais. Will they be able to manage to pay the bill within the amount they had?

Cramer's rule

(OR)

b. If $\vec{a} = \hat{\imath} - \hat{\jmath}, \vec{b} = \hat{\imath} - \hat{\jmath} - 4\hat{k}, \vec{c} = 3\hat{\jmath} - \hat{k}$ and $\vec{d} = 2\hat{\imath} + 5\hat{\jmath} + \hat{k}$, verify that

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{b}, \vec{d}]\vec{c} - [\vec{a}, \vec{b}, \vec{c}]\vec{d}$$

42. a. If $z = x + iy$ is a complex number such that $Im\left(\frac{2z+1}{iz+1}\right) = 0$, show that $2x^2 + 2y^2 + x - 2y = 0$.

(OR)

b. Prove by vector method that the perpendiculars (altitudes) from the vertices to the opposite sides of a triangle are concurrent.

43. a. If $cos^{-1} x + cos^{-1} y + cos^{-1} z = \pi$ and $0 < x, y, z < 1$, then show that $x^2 + y^2 + z^2 + 2xyz = 1$.

(OR)

b. Find solution, if any, of the equation $2\,cos^2 x - 9\,cos\,x + 4 = 0$.

44. a. At a water fountain, water attains a maximum height of 4m at horizontal distance of 0.5m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75m from the point of origin

0.75×0.75

(OR)

b. Find all cube roots of $\sqrt{3} + i$.

45. a. Find the centre, foci and eccentricity of the hyperbola $11x^2 - 25y^2 - 44x + 50y - 256 = 0$.

(OR)

b. Find the foot of the perpendicular drawn from the point (5,4,2) to the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$. Also, find the equation of the perpendicular.

0.75×100

46. a. Find the value of $cos\left(sin^{-1}\left(\frac{4}{5}\right) - tan^{-1}\left(\frac{3}{4}\right)\right)$.

(OR)

b. Investigate for what values of λ and μ the system of linear equations

$x + 2y + z = 7, x + y + \lambda z = \mu, x + 3y - 5z = 5$ has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

47. a. If the line $y = mx + c$ is a common tangent to the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ and the circle $x^2 + y^2 = 36$ then prove that $4c^2 = 369$

(OR)

b. Solve the equation $15x^4 - 8x^3 - 14x^2 + 8x - 1 = 0$ given that the roots are in H.P